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Waterproof transformer coil structure and manufacturing method thereof

Abstract

Disclosed are a waterproof transformer coil structure and a manufacturing method thereof. The waterproof transformer coil structure includes: an insulating cylinder configured for supporting a transformer coil; a strut configured for being wound with the transformer coil, where the strut is mounted on the insulating cylinder, and the strut is provided with strip-shaped notches; and a winding wire, where the winding wire is wound on the strip-shaped notches.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5416373	12/1994	Maruyama et al.	N/A	N/A
2007/0279177	12/2006	Sarver	29/605	H01F 41/063
2015/0376672	12/2014	Shin et al.	N/A	N/A
2016/0027573	12/2015	Xu et al.	N/A	N/A
2017/0352473	12/2016	Wang	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
204117812	12/2014	CN	N/A
205248106	12/2015	CN	N/A
206022077	12/2016	CN	N/A
108847330	12/2017	CN	N/A
112786293	12/2020	CN	N/A
214279765	12/2020	CN	N/A
102014116973	12/2015	DE	N/A
S63188911	12/1987	JP	N/A
2006147927	12/2005	JP	N/A
2011204836	12/2010	JP	N/A
2016032094	12/2015	JP	N/A
6028109	12/2015	JP	N/A
2014017213	12/2013	WO	N/A

OTHER PUBLICATIONS

English translation of CN206022077 (Year: 2017). cited by examiner

First Office Action for the Chinese family patent CN112786293A on Nov. 29, 2024 with English translation. cited by applicant

First Office Action for the German family patent DE112021000113.2 on Aug. 30, 2024 with

English translation. cited by applicant

First Office Action for the Indian family patent No. 202227014369 on Aug. 9, 2024. cited by applicant

First Office Action for the Japanese family patent JP2023520271A on May 16, 2023. cited by applicant

First Search Report for the Chinese family patent CN112786293A on Nov. 28, 2024 with English translation. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage filing under 35 U.S.C. § 371 of international application number PCT/CN2021/129881, filed Nov. 10, 2021, which claims priority to Chinese patent application No. 2021100177092 filed Jan. 7, 2021. The contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The disclosure relates to the field of transformers, and more particularly, to a waterproof transformer coil structure and a manufacturing method thereof.

BACKGROUND

(3) In recent years, underground distribution devices may be flooded and burned down due to waterlogging, which leads to interruption of power supply. Once the transformer, as the key component of a distribution device, is burned down, on-site replacement of the transformer is difficult and takes a long time, which will cause long-term power outage, seriously reduce an efficiency of power supply restoration of an urban power grid and cause great inconvenience to people's life and production. Considering aspects such as safety, reliability and fireproofing, dry-type transformer is generally used as an underground transformer. Performance of the coil, as a core member of the transformer, directly affects whether the product can be operated normally. However, when the dry-type transformer undergoes waterlogging, the coil, after soaked during waterlogging, will be damp and has a reduced electrical performance or even be burnt-down, thus affecting normal operation of the product. For a traditional waterproof transformer, the coil is casted into a whole by using epoxy casting, which is complicated in process, and has high material and production costs. When the product is operated for a period of time, the coil is prone to a "crazing" phenomenon, which reduces an electrical performance and a waterproof capability of the product.

SUMMARY

(4) The disclosure aims to solve at least one of the technical problems in the existing technology. To this end, the disclosure provides a waterproof transformer coil structure and a manufacturing method thereof. The waterproof transformer coil structure solves problems that a coil is easy to get damp and an electrical performance is lowered after a dry-type transformer is waterlogged, and the manufacturing method is provided, so that technical effects of a simple and reliable operation and improvement of a cost performance of products are achieved.

(5) The disclosure provides a waterproof transformer coil structure with the functions above. The waterproof transformer coil structure includes: an insulating cylinder configured for supporting a

transformer coil; a strut configured for being wound with the transformer coil, where the strut is mounted on the insulating cylinder, and the strut is provided with a plurality of strip-shaped notches; and a winding wire, where the winding wire is wound on the strip-shaped notches.

(6) The waterproof transformer coil structure according to the embodiments of the disclosure at least has the following technical effects: the waterproof transformer coil structure improves waterproof and water drainage capabilities of the coil and enhances a waterproof performance of the transformer through the strut and the strip-shaped notches arranged in the strut.

(7) For the waterproof transformer coil structure according to the embodiments of the disclosure, the strut is provided in plural, and the struts are evenly arranged on an outer wall of the insulating cylinder, so that the struts are evenly distributed on the insulating cylinder. The struts can enhance a water drainage capability of the waterproof transformer coil, water can flow out of the coil along the struts without water accumulation, and the even distribution of the struts on the insulating cylinder can enhance a mechanical strength of the insulating cylinder.

(8) For the waterproof transformer coil structure according to the embodiments of the disclosure, the strip-shaped notches are evenly arranged on an outer edge of the strut, so that the strip-shaped notches are evenly distributed on the strut. The even distribution of the strip-shaped notches on the strut can enhance a mechanical strength of the strut, and the winding wire is wound on the strip-shaped notches, so that the even distribution of the strip-shaped notches can maintain an electrical performance of the transformer.

(9) For the waterproof transformer coil structure according to the embodiments of the disclosure, two ends of the strut are respectively provided with a bending notch which is convenient for water drainage. An angle ring is inserted into the bending notch, which can enhance an electrical performance and reduce accumulated water in the coil.

(10) For the waterproof transformer coil structure according to the embodiments of the disclosure, the bending notch includes an inclined portion convenient for water drainage and a vertical portion, and the inclined portion is connected with the vertical portion. The inclined portion is convenient for water drainage, and after the angle ring is inserted into the vertical portion, the electrical performance can be enhanced and the accumulated water in the coil can be reduced.

(11) The waterproof transformer coil structure according to the embodiments of the disclosure further includes an angle ring, where an outer end face of the angle ring is lower than an inner end face of the angle ring, the angle ring is arranged on the bending notch of the strut, and the angle ring is made of silicone rubber material. Arrangement of the outer end face of the angle ring being lower than the inner end face of the angle ring can facilitate outflow of the water, and reduce a probability of water accumulation in the transformer coil.

(12) For the waterproof transformer coil structure according to the embodiments of the disclosure, the winding wire includes a metal wire conductor, a first polyimide film, a first insulating paper, a second insulating paper and a second polyimide film. The metal wire conductor, the first polyimide film, the first insulating paper, the second insulating paper and the second polyimide film are wrapped in sequence, and the winding wire is wound on the strip-shaped notches. Wrapping of the metal wire conductor, the first polyimide film, the first insulating paper, the second insulating paper and the second polyimide film in sequence can enhance insulation of the metal wire conductor, and prevent short circuit caused by water soaking.

(13) The waterproof transformer coil structure according to the embodiments of the disclosure further includes an insulating ring arranged on both end faces of the winding wire, and the insulating ring is made of an insulating material. The insulating rings are used for enhancing an insulating performance of the transformer coil.

(14) A method for manufacturing the waterproof transformer coil according to the embodiments of the disclosure includes: machining an insulating material or an insulating plate into a cylindrical structure to form an insulating cylinder; machining an insulating material into a plurality of struts, and machining a plurality of strip-shaped notches in a middle portion of each of the struts; evenly

distributing and fixing the struts on a surface of the insulating cylinder; and winding a winding wire on the strip-shaped notches of the strut to form a pie winding.

(15) The method for manufacturing the waterproof transformer coil according to the embodiments of the disclosure further: inserting an insulating ring on both upper and lower end faces of the pie winding; inserting an angle ring along a bending notch of the strut; performing primary roasting on the waterproof transformer coil; performing primary vacuum immersion paint on the waterproof transformer coil; performing secondary roasting on the waterproof transformer coil; performing secondary vacuum immersion paint on the waterproof transformer coil; performing third roasting on the waterproof transformer coil; and coating hydrophobic paint on a surface of the waterproof transformer coil.

(16) The additional aspects and advantages of the disclosure will be given in part in the following description, and will become apparent in part from the following description, or will be learned through the practice of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and/or additional aspects and advantages of the disclosure will be apparent and easily understood from the description of the embodiments with reference to the following accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram of a waterproof transformer coil structure according to the embodiments of the disclosure;

(3) FIG. 2 is a schematic diagram of a strut of the waterproof transformer coil structure according to the embodiments of the disclosure;

(4) FIG. 3 is a schematic diagram of a winding wire of the waterproof transformer coil structure according to the embodiments of the disclosure;

(5) FIG. 4 is a schematic diagram of an angle ring of the waterproof transformer coil structure according to the embodiments of the disclosure; and

(6) FIG. 5 is a schematic diagram of an insulating ring of the waterproof transformer coil structure according to the embodiments of the disclosure.

REFERENCE NUMERALS

(7) **101** refers to insulating cylinder, **201** refers to strut, **202** refers to strip-shaped notch, **230** refers to bending notch, **204** refers to inclined portion, **205** refers to vertical portion, **301** refers to winding wire, **302** refers to metal wire conductor, **303** refers to first polyimide film, **304** refers to first insulating paper, **305** refers to second insulating paper, **306** refers to second polyimide film, **401** refers to angle ring, and **501** refers to insulating ring.

DETAILED DESCRIPTION

(8) The embodiments of the disclosure are described below in detail. Examples of the embodiments are shown in the accompanying drawings. The same or similar numerals represent the same or similar elements or elements having the same or similar functions throughout the specification. The embodiments described below with reference to the accompanying drawings are exemplary, and are only used to explain the disclosure but should not be construed as a limitation to the disclosure.

(9) In the description of the disclosure, it should be understood that the positional descriptions referred to, for example, the directional or positional relationships indicated by up, down, front, rear, left, right, etc., are based on the directional or positional relationships shown in the drawings, and are only for convenience and simplification of description of the disclosure, but not for indicating or implying that the referred device or element must have a specific direction, be constructed and operated in a specific direction, and thus should not be construed as limiting the disclosure.

(10) In the description of the disclosure, “several” means one or more, “a plurality of” means more than two, “greater than a number”, “less than a number”, “exceed a number” and the like indicate that the number is excluded, and “above a number”, “below a number”, “within a number”, and the like indicate that the number is included. “First” and “second” are only used to distinguish between technical features but cannot be used to indicate or imply relative importance or implicitly specify a quantity of indicated technical features or implicitly specify a sequential relationship of indicated technical features.

(11) In the description of the disclosure, unless otherwise expressly defined, the terms such as “disposed”, “mounted”, and “connected” should be understood in a broad sense. For persons of ordinary skill in the art, specific meanings of the terms in the disclosure may be appropriately determined with reference to the specific content in the technical solution.

(12) With reference to FIG. 1, the disclosure provides a waterproof transformer coil structure with the functions described above. The waterproof transformer coil structure includes: an insulating cylinder **101** configured for supporting a transformer coil; a strut **201** configured for being wound with the transformer coil, where the strut **201** is mounted on the insulating cylinder **101**, and the strut **201** is provided with strip-shaped notches **202**; and a winding wire **301**, where the winding wire **301** is wound on the strip-shaped notches **202**.

(13) With reference to FIG. 1, the waterproof transformer coil structure according to the embodiments of the disclosure at least has the following technical effects: the waterproof transformer coil structure improves waterproof and water drainage capabilities of the coil and enhances a waterproof performance of the transformer through the strut **201** and the strip-shaped notches **202** arranged in the strut **201**.

(14) With reference to FIG. 2, for the waterproof transformer coil structure according to the embodiments of the disclosure, a plurality of struts **201** are provided, and the struts **201** are evenly arranged on an outer wall of the insulating cylinder **101**, so that the struts **201** are evenly distributed on the insulating cylinder **101**. The struts **201** can enhance a water drainage capability of the waterproof transformer coil, water can flow out of the coil along the struts **201** without water accumulation, and even distribution of the struts **201** on the insulating cylinder **101** can enhance a mechanical strength of the insulating cylinder **101**.

(15) With reference to FIG. 2, for the waterproof transformer coil structure according to the embodiments of the disclosure, the strip-shaped notches **202** are evenly arranged on an outer edge of the strut **201**, so that the strip-shaped notches **202** are evenly distributed on the strut **201**. The even distribution of the strip-shaped notches **202** on the strut **201** can enhance a mechanical strength of the strut **201**, and the winding wire **301** is wound on the strip-shaped notches **202**, so that the even distribution of the strip-shaped notches **202** can maintain an electrical performance of the transformer.

(16) With reference to FIG. 2, for the waterproof transformer coil structure according to the embodiments of the disclosure, two ends of the strut **201** are respectively provided with a bending notch **203** which is convenient for water drainage. An angle ring **401** is inserted into the bending notch **203**, which can enhance an electrical performance and reduce accumulated water in the coil.

(17) With reference to FIG. 2, for the waterproof transformer coil structure according to the embodiments of the disclosure, the bending notch **203** includes an inclined portion **204** convenient for water drainage and a vertical portion **205**, and the inclined portion **204** is connected with the vertical portion **205**. The inclined portion **204** is convenient for water drainage, and after the angle ring **401** inserted into the vertical portion **205**, the electrical performance can be enhanced and the accumulated water in the coil can be reduced.

(18) With reference to FIG. 1 and FIG. 4, the waterproof transformer coil structure according to the embodiments of the disclosure further includes an angle ring **401**, where an outer end face of the angle ring **401** is lower than an inner end face of the angle ring **401**, the angle ring **401** is arranged in the bending notch **203** of the strut **201**, and the angle ring **401** is made of a silicone rubber

material. Arrangement of the outer end face of the angle ring **401** being lower than the inner end face of the angle ring **401** can facilitate outflow of the water, and reduce a probability of water accumulation in the transformer coil.

(19) With reference to FIG. **1** and FIG. **3**, for the waterproof transformer coil structure according to the embodiments of the disclosure, the winding wire **301** includes a metal wire conductor **302**, a first polyimide film **303**, a first insulating paper **304**, a second insulating paper **305** and a second polyimide film **306**. The metal wire conductor **302**, the first polyimide film **303**, the first insulating paper **304**, the second insulating paper **305** and the second polyimide film **306** are wrapped in sequence, and the winding wire **301** is wound on the strip-shaped notches **202**. Wrapping of the metal wire conductor **302**, the first polyimide film **303**, the first insulating paper **304**, the second insulating paper **305** and the second polyimide film **306** in sequence can enhance insulation of the metal wire conductor **302**, and prevent short circuit caused by water soaking.

(20) With reference to FIG. **1** and FIG. **5**, the waterproof transformer coil structure according to the embodiments of the disclosure further includes an insulating ring **501** on both end faces of the winding wire **301**, and the insulating ring **501** is made of insulating material. The insulating ring **501** is used for enhancing an insulating performance of the transformer coil.

(21) With reference to FIG. **1**, a method for manufacturing the waterproof transformer coil according to the embodiments of the disclosure includes the following steps of: machining an insulating material or an insulating plate into a cylindrical structure to form an insulating cylinder **101**; machining an insulating material into struts **201**, and machining strip-shaped notches **202** in a middle portion of each of the struts **201**; evenly distributing and fixing the struts **201** on a surface of the insulating cylinder **101**; and winding the winding wire **301** on the strip-shaped notches **202** of the struts **201** to form a pie winding.

(22) With reference to FIG. **1**, the method for manufacturing the waterproof transformer coil according to the embodiments of the disclosure further includes the steps of: inserting an insulating ring **501** on both upper and lower end faces of the pie winding; inserting an angle ring **401** along a bending notch **203** of each of the struts **201**; performing primary roasting on the waterproof transformer coil; performing primary vacuum immersion paint on the waterproof transformer coil; performing secondary roasting on the waterproof transformer coil; performing secondary vacuum immersion paint on the waterproof transformer coil; performing third roasting on the waterproof transformer coil; and coating hydrophobic paint on a surface of the waterproof transformer coil.

(23) The embodiments of the disclosure are described in detail with reference to the accompanying drawings above, but the disclosure is not limited to the above embodiments, and various changes may also be made within the knowledge scope of those of ordinary skills in the art without departing from the purpose of the disclosure.

(24) In the description of the disclosure, the descriptions referring to the terms “one embodiment”, “some embodiments”, “schematic embodiments”, “examples”, “specific examples”, or “some examples” mean that the specific features, structures, materials, or characteristics described in combination with the embodiment or example are included in at least one embodiment or example of the disclosure. In the specification, the schematic representation of the above terms does not necessarily mean the same embodiment or example. Moreover, the specific features, structures, materials or characteristics described may be combined in any one or more embodiments or examples in a suitable manner.

(25) Although the embodiments of the disclosure have been shown and described, those of ordinary skills in the art may understand that various changes, modifications, substitutions and variations may be made to these embodiments without departing from the principle and purpose of the disclosure, and the scope of the disclosure is defined by the claims and their equivalents.

Claims

1. A waterproof transformer coil structure, comprising: an insulating cylinder configured for supporting a transformer coil; a strut configured for being wound with the transformer coil, wherein the strut is mounted on the insulating cylinder, and the strut is provided with strip-shaped notches; and a winding wire, wherein the winding wire is wound on the strip-shaped notches, wherein two ends of the strut are respectively provided with a bending notch for water drainage, and wherein the bending notch comprises an inclined portion for water drainage and a vertical portion, and the inclined portion is connected with the vertical portion.
 2. The waterproof transformer coil structure of claim 1, wherein the strut is provided in plural, and the struts are evenly arranged on an outer wall of the insulating cylinder.
 3. The waterproof transformer coil structure of claim 1, wherein the strip-shaped notches are evenly arranged on an outer edge of the strut.
 4. The waterproof transformer coil structure of claim 1, further comprising an angle ring, wherein an outer end face of the angle ring is lower than an inner end face of the angle ring, and the angle ring is arranged on the bending notch of the strut.
 5. The waterproof transformer coil structure of claim 1, wherein the winding wire comprises a metal wire conductor, a first polyimide film, a first insulating paper, a second insulating paper and a second polyimide film; the metal wire conductor, the first polyimide film, the first insulating paper, the second insulating paper and the second polyimide film are wrapped in sequence, and the winding wire is wound on the strip-shaped notches.
 6. The waterproof transformer coil structure of claim 1, further comprising an insulating ring arranged on both end faces of the winding wire.
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